

Week 11 Design Stage Report

PTAT Current Reference Hand-Calculations

1. Design Objectives

- **Target Current (I_{PTAT}):** 10 μ A
- **Target Temperature Coefficient (TC):** $\sim +3300$ ppm/ $^{\circ}$ C
- **Topology:** Diode-connected MOSFETs (weak inversion) or parasitic BJTs

2. Core Theoretical Equations

The PTAT topology generates a current based on the difference in the base-emitter (or gate-source in weak inversion) voltages of two devices with an area ratio of N.

- **Voltage Difference (ΔV_{BE}):** $\Delta V_{BE} = V_T * \ln(N) = (kT/q) * \ln(N)$
- **Current Output (I_{PTAT}):** $I_{PTAT} = \Delta V_{BE} / R = (V_T * \ln(N)) / R$

3. Area Ratio and Resistor Sizing

To optimize common-centroid layout matching around a single center device, a standard area ratio of 8 is selected.

- **Selected Area Ratio (N):** 8
- **Thermal Voltage (V_T) at 300K (Room Temperature):** ~ 25.85 mV
- **Calculated ΔV_{BE} :**
 - $\Delta V_{BE} = 25.85 \text{ mV} * \ln(8) = 53.75 \text{ mV}$
- **Calculated Resistor (R) for 10 μ A Target:**
 - $R = 53.75 \text{ mV} / 10 \mu\text{A} = 5.375 \text{ k}\Omega$

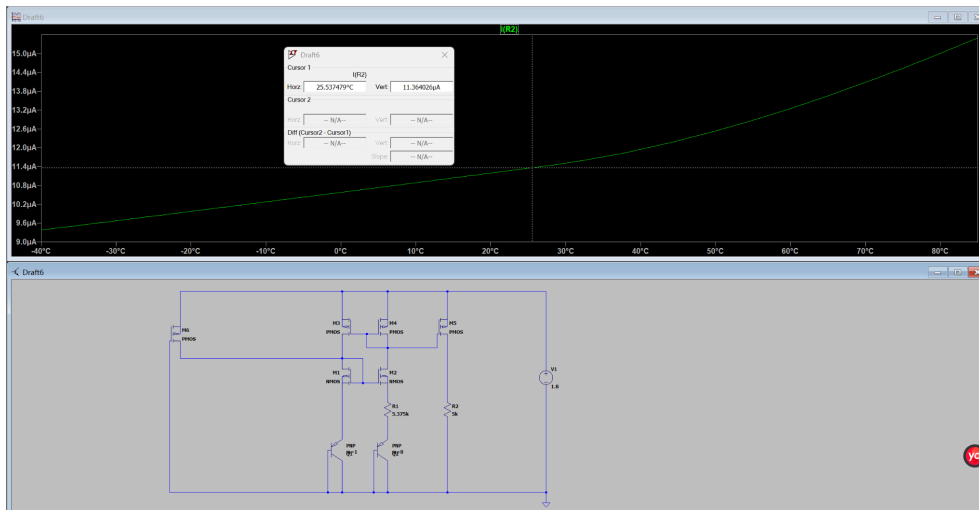
4. Temperature Coefficient (TC) Verification

To ensure the design meets the target TC of $\sim +3300$ ppm/ $^{\circ}$ C, the theoretical derivative of I_{PTAT} with respect to temperature (T) is calculated, assuming the temperature coefficient of resistor R is ideally zero.

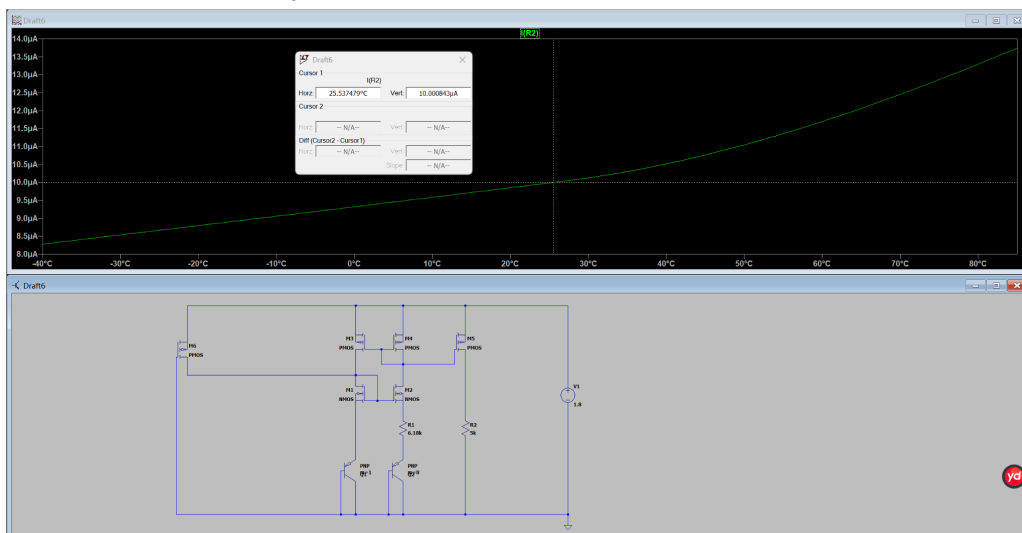
- **Partial Derivative:** $\partial(I_{PTAT})/\partial T = (k / qR) * \ln(N) = I_{PTAT} / T$
- **TC Equation:** $TC = (1 / I_{PTAT}) * (\partial(I_{PTAT})/\partial T) = 1 / T$
- **Calculated TC at 300K:**
 - $TC = 1 / 300 = 0.003333 = +3333 \text{ ppm}/^{\circ}\text{C}$

5. Simulation

Theoretical Value (Hand-Calculated): Based on first-order ideal equations with an area ratio of N=8, the required resistor was calculated as $R_1 = 5.375 \text{ k}\Omega$.



Simulated & Tuned Value: To precisely achieve the 10 μ A target at 25°C in LTspice, the resistor was empirically tuned to $R1 = 6.18 \text{ k}\Omega$



The deviation between the ideal hand-calculated value and the required SPICE parameter stems from three primary physical non-idealities:

- **Startup Circuit Leakage:** The continuous-conduction startup PMOS, despite utilizing a long-channel geometry to restrict flow, still injects a nano-ampere level parasitic current into the core loop. This elevates the overall baseline current.
- **Channel-Length Modulation:** The V_{DS} across the basic current mirror pairs are unequal. This causes a mismatch in the current mirroring ratio, breaking the ideal 1:1 assumption used in hand calculations.
- **BJT Model Non-Idealities:** The SPICE PNP models account for parasitic series resistances and base current losses, causing the actual ΔV_{BE} to deviate slightly from the ideal theoretical model.

CS/cascode Amplifier

1. Design Objective

- **Voltage Gain (A_v):** $\geq 36 \text{ dB}$

- **Bandwidth (BW):** ≥ 100 kHz (with $C_L = 5$ pF)
- **Input-Referred Noise:** < 15 nV/ $\sqrt{\text{Hz}}$ at 1 kHz
- **Output Swing:** > 800 mVpp

2. Theoretical Analysis and Hand-Calculations

Noise Constraint The dominant thermal noise contribution comes from the input MOSFET. The required minimum transconductance (g_{m1}) is derived from the power spectral density formula: $V_{n,in}^2 = (4 * k * T * \gamma) / g_{m1}$

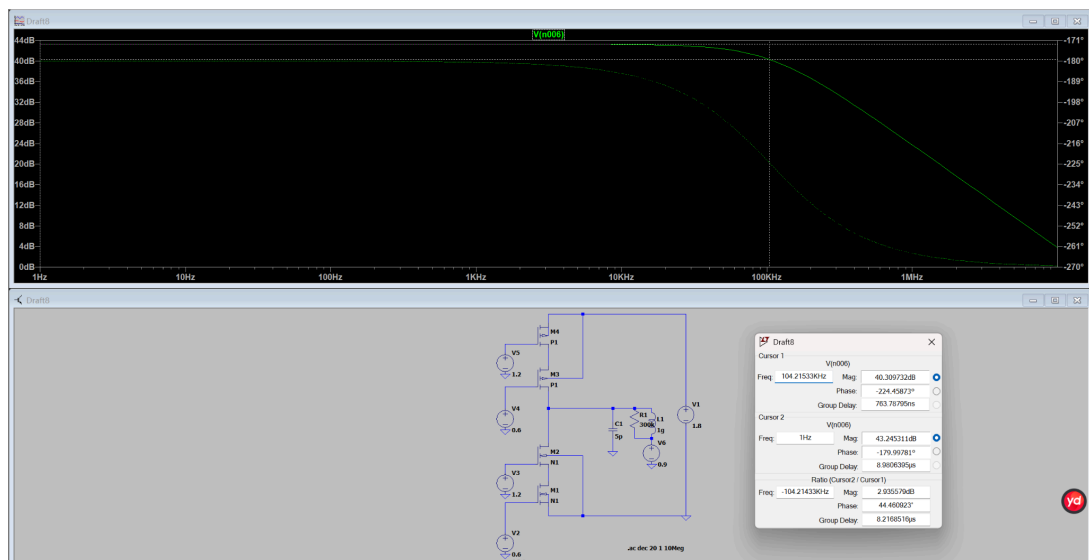
Assuming $T = 300$ K and $\gamma \approx 2/3$, to achieve $V_{n,in} < 15$ nV/ $\sqrt{\text{Hz}}$, the theoretical minimum requirement is **$g_{m1} \geq 49$ μS** .

Bandwidth Constraint To maintain the dominant pole at or above 100 kHz with a 5 pF load capacitance, the maximum allowable output impedance (R_{out}) is calculated as: $R_{out} \leq 1 / (2 * \pi * f_{3dB} * C_L) \approx 318$ k Ω

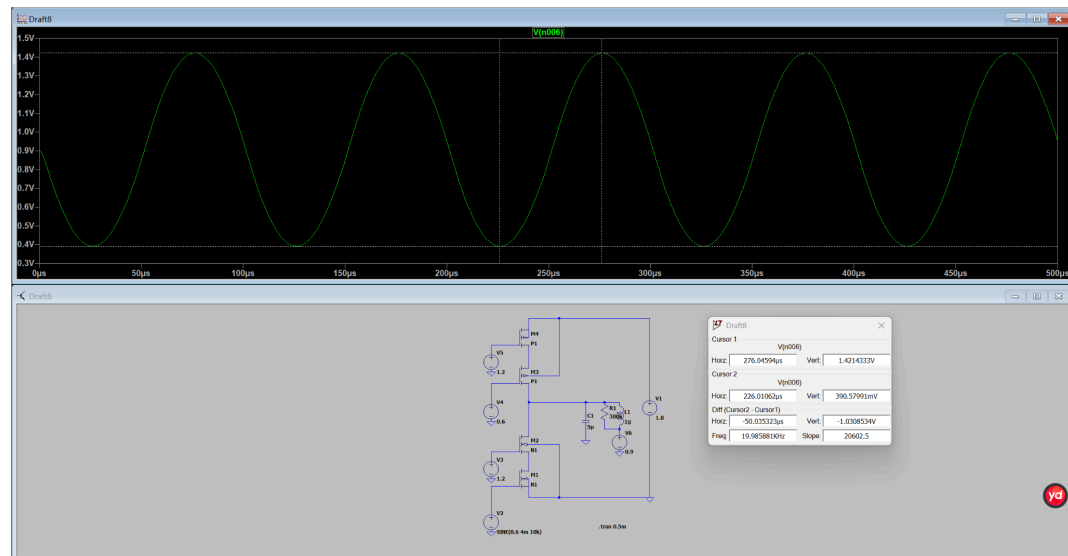
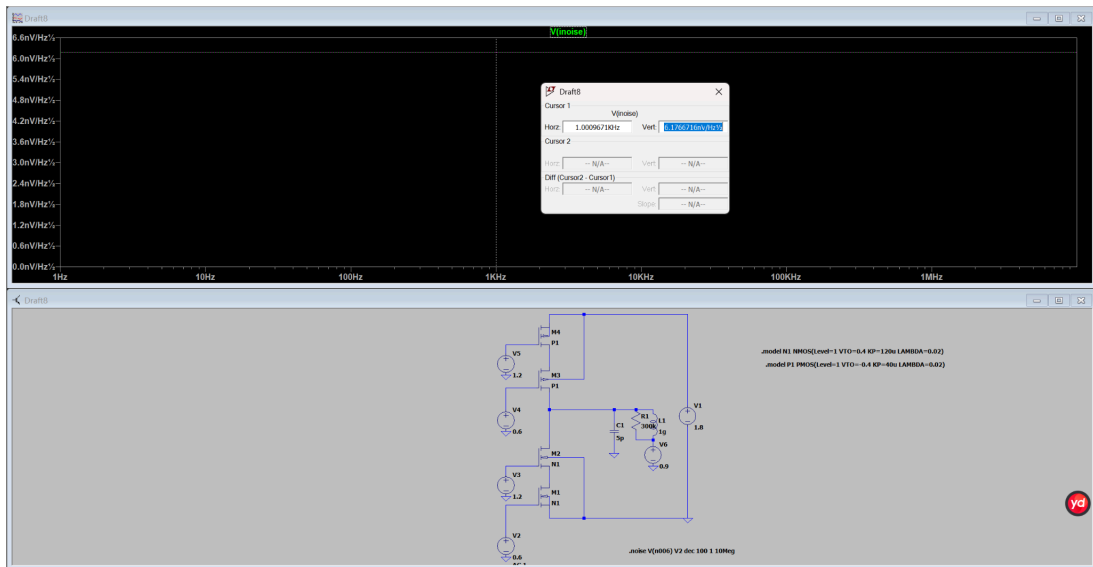
Gain-Bandwidth Synthesis To achieve a DC gain of 36 dB (approximately 63.1 V/V) while strictly adhering to the 318 k Ω output impedance limit, the required target g_{m1} must be increased to accommodate the gain-bandwidth product (GBW): $g_{m1} = A_v / R_{out} \approx 198$ μS

3. Simulation Results

Performance Summary



Voltage Gain (A_v): 43 dB Bandwidth (BW): 100 kHz



Detailed Verification Notes

- DC Operating Point:** All four MOSFETs operate securely within the saturation region. The extracted actual transconductance of the input device is **gm1 = 486 μS**. This massive margin over the 49 μS theoretical minimum is the primary reason the thermal noise is heavily suppressed to an excellent 6.18 nV/√Hz.
- Transient Analysis (Output Swing):** With a 4 mV amplitude, 10 kHz input sine wave, the output demonstrates a clean, undistorted swing from 390 mV to 1.42 V. This yields a peak-to-peak voltage of **1.03 Vpp**, satisfying the large-signal requirement without clipping.
- AC Analysis:** The parallel 300 kΩ load successfully trades excess intrinsic cascode gain for the required 100 kHz bandwidth, securing an optimal GBW utilization.

Two-Stage Fully Differential Folded-Cascode OTA

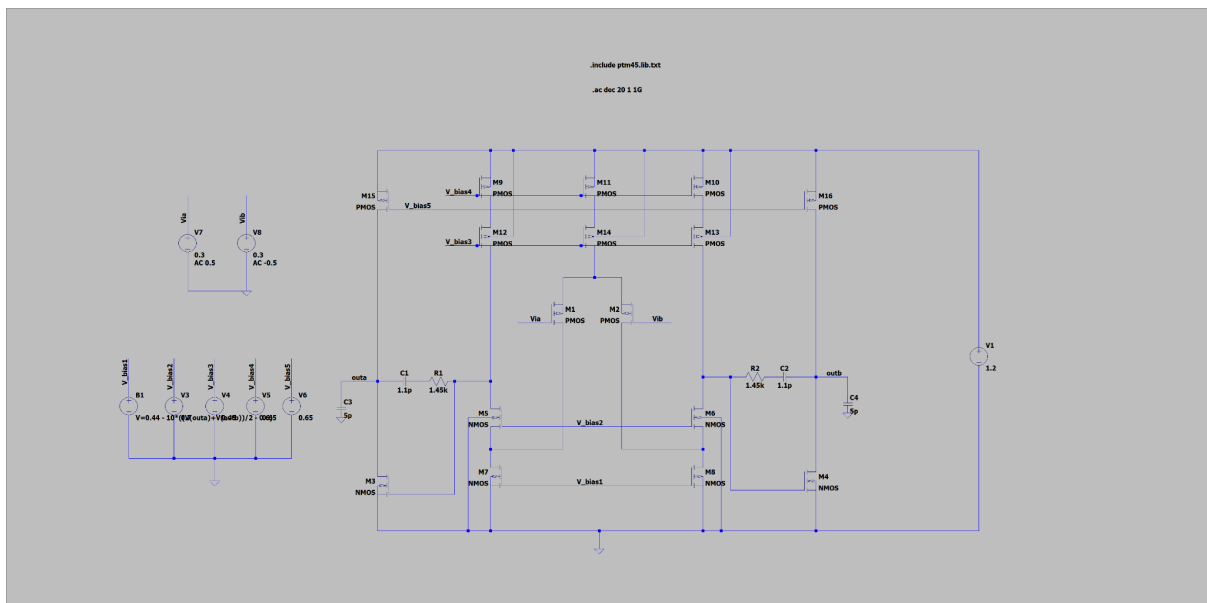
1. Design Objectives

This report details the design and simulation of a two-stage fully differential operational transconductance amplifier (OTA). The architecture consists of a PMOS-input folded-cascode first stage and a common-source second stage, utilizing a 45nm CMOS process with a 1.2V supply voltage.

The primary design objectives for the OTA with a 5 pF capacitive load are:

- DC Gain: > 80 dB
- Unity Gain Frequency (UGF / GBW): > 10 MHz
- Phase Margin (PM): > 60°

2. Circuit Architecture and Device Sizing



To achieve the high gain requirement while avoiding short-channel effects, non-minimum channel lengths ($L = 200$ nm) were selected for all transistors.

- First Stage (Folded-Cascode): A PMOS input differential pair is used to accommodate a lower input common-mode voltage ($V_{icm} = 0.3$ V). The folded-cascode topology provides high output impedance and significant voltage gain. Multiplier (M) parameters are heavily utilized in the layout-matching strategy to mirror precise current ratios.
- Second Stage (Common-Source): High-drive-strength PMOS and NMOS transistors ($W = 10$ μ m, $M = 5$) are implemented to push the non-dominant pole to a higher frequency and satisfy the phase margin requirement while driving the large 5 pF load.
- Compensation Network: Standard Miller compensation is implemented utilizing a capacitor (C_c) and a nulling resistor (R_c) in series across the second stage to eliminate the right-half-plane (RHP) zero.

3. Key Design Equations

Total DC Gain (A_v) The overall low-frequency open-loop gain is the product of the first and second stage gains: $A_v = (g_{m1} * R_{out1}) * (g_{m2} * R_{out2})$

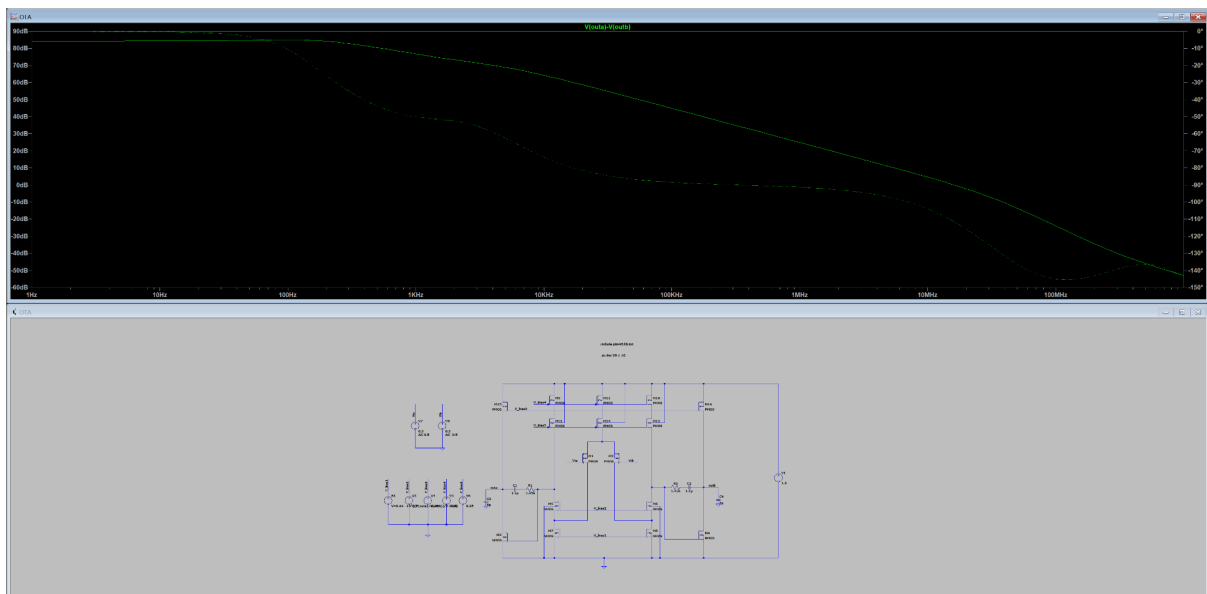
Unity Gain Frequency (GBW) The gain-bandwidth product is primarily determined by the transconductance of the input pair and the Miller compensation capacitor: $GBW = g_{m1} / (2 * \pi * C_c)$

Slew Rate (SR) The maximum rate of change of the output voltage is dictated by the tail current of the first stage charging the compensation capacitor: $SR = I_{tail} / C_c$

Non-Dominant Pole (p_2) To maintain stability, the second pole must be pushed beyond the UGF. It is dictated by the transconductance of the second stage and the load capacitor: $p_2 = g_{m2} / (2 * \pi * C_L)$

Nulling Resistor for RHP Zero (R_c) The right-half-plane zero introduced by the Miller capacitor is pushed to infinity by matching the nulling resistor to the inverse of the second stage transconductance: $R_c = 1 / g_{m2}$

4. Final Component Parameters



Based on the theoretical calculations and iterative LTspice operating point (.op) optimization, the final compensation parameters are:

- Load Capacitance (C_L): 5 pF
- Miller Capacitance (C_1, C_2): 1.1 pF
- Nulling Resistor (R_1, R_2): 1.45 k Ω
- Ideal CMFB Target: 0.6 V

5. Simulation Results Summary

Note on CMFB Implementation: To isolate the potential secondary pole interference from a physical common-mode feedback loop and accurately verify the AC limit characteristics of

the core OTA itself, an ideal behavioral voltage source (ideal CMFB) was implemented for DC locking in this verification phase.

DC Operating Point Analysis:

- All transistors are successfully biased in the saturation region ($|V_{ds}| > |V_{dsat}|$).
- The output common-mode voltage is perfectly locked at 0.6 V, confirming the efficacy of the feedback loop.

AC Analysis (Bode Plot):

- DC Gain: ≈ 84 dB (Exceeds the 80 dB target)
- GBW: ≈ 25 MHz (Exceeds the 10 MHz target)
- Phase Margin: $\approx 70^\circ$ (Exceeds the 60° target)